

Revised Generation Interconnection System Impact Study Report

For Queue Project Q16-0636

Speed 138 kV
November 22, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0636. The Interconnection Customer proposes a 65.8 MW natural gas combined cycle facility interconnecting at the Speed 138 kV substation in the Kentucky area. The request was received on June 25, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 270 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,287,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
BEANBLOSSOM-SW STA – GREENVILLE AVE 138.0 kV	LEE COUNTY SOLID WASTE ENERGY – LIVERMORE 345.0 kV	121.8	343	6.9
RANCHO CARMEL – RUSSELLVILLE EAST 138.0 kV	ALBION RD SUBSTATION – PLATTE CITY 138.0 kV	111.8	266	7.5
SAND SPRINGS – RUSSELLVILLE EAST 138.0 kV	ALBION ROAD – PLATTE CITY 138.0 kV	109.3	265	7.5
ALIBATS – JULIETTE 138.0 kV line	PINE BLUFF WATSON CHAPEL – CARRAIZO 1201 345.0 kV	108.1	296	26.9
ALIBATES – LATAH JCT 138.0 kV	LIVERMORE – LEE COUNTY SOLID WASTE ENERGY 345.0 kV	107.5	322	5.9
SILVER BAY POWER – DAMON 138.0 kV	SILVER BAY POWER – HOTEL ROAD 138.0 kV	104.4	192	33.9
WAGON MOUND – UNIVERSITY 138.0 kV	PINE BLUFF WATSON CHAPEL – CARRAIZO 1201 345.0 kV	104.3	227	16.1
SANTA ROSA BEACH – TULE 138.0 kV line	DAISVILLE – N APPLETON 138.0 kV line	103.8	148	5.9
SPATSUM LORNEX – LATAH JCT 138.0 kV	LIVERMORE – LEE COUNTY SOLID WASTE ENERGY 345.0 kV Line	102.5	349	5.9
ELDORADO – ACTION 138.0 kV	PINE BLUFF WATSON CHAPEL – CARRAIZO 1201 345.0 kV	101.1	548	32.8
ALIBATES – ACTION 138.0 kV Line	PINE BLUFF WATSON CHAPEL – CARRAIZO 1201 345.0 kV line	101.0	587	32.8
WELLINGTON – DAVISVILLE 138.0 kV L	ALBION ROAD – PLATTE CITY 138.0 kV	101.0	396	6.7

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
RICHARDSON – CLEVELAND 138.0 kV	RICHARDSON SUBSTATION – CLEVELAND 138.0 kV Line 2	100.5	284	6.9
RICHARDSON – CLEVELAND 138.0 kV	RICHARDSON – CLEVELAND 138.0 kV #1	100.5	284	6.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BEANBLOSSOM-SW STA – GREENVILLE AVE 138.0 kV	Rebuild line with higher-capacity conductor	\$24,893,000	2% (\$495,000)
RACHO CARMEL – RUSSELLVILLE E 138.0 kV	Rebuild line with higher-capacity conductor	\$9,711,000	3% (\$254,000)
SAND SPRINGS – RUSSELLVILLE EAST 138.0 kV	Reconductor line	\$2,806,000	3% (\$73,000)
ALIBATS – JULIETTE 138.0 kV line	Reconductor line	\$4,130,000	6% (\$240,000)
ALIBATES – LATAH JCT 138.0 kV	Reconductor line	\$3,932,000	2% (\$94,000)
SILVER BAY POWER – DAMON 138.0 kV	Reconductor line	\$1,237,000	8% (\$97,000)
WAGON MOUND – UNIVERSITY 138.0 kV	Reconductor line	\$4,503,000	6% (\$291,000)
SANTA ROSA BEACH – TULE SUBSTATION 138.0 kV	Reconductor line	\$1,884,000	2% (\$44,000)
SPATSUM LORNEX – LATAH JUNCTION 138.0 kV Line	Reconductor line	\$2,118,000	2% (\$50,000)
ELDORADO – ACTION 138.0 kV	Reconductor line	\$888,000	5% (\$42,000)
ACTION – ALIBATES 138.0 kV L	Reconductor line	\$9,170,000	5% (\$431,000)
DAVISVILLE – WELLINGTON 138.0 kV Line	Reconductor line	\$1,386,000	2% (\$22,000)
RICHARDSON – CLEVELAND 138.0 kV Cir. 1	Reconductor line	\$1,955,000	4% (\$88,000)
RICHARDSON – CLEVELAND 138.0 kV	Reconductor line	\$1,480,000	4% (\$66,000)
	Total allocated to project		\$2,287,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.